GenAI Quality and Maturity Assessment

Beyond the Demo

SYSTEM	TEAM
Awesome GenAI system	**Awesome Team**
DATE	ASSESSOR
2026-03-14 09:11	**Joe Doe**

QUALITY SCORE	ACTUAL MATURITY
58.00	**L1**
REQUIRED MATURITY	CRITICALITY
L3	**Production Non Critical**

Below required maturity **Needs major quality improvements**

Maturity Level

L1 L2 L3 L4 L5

Current maturity Expected maturity based on business criticality

Characteristic Scores

Utility

75.0

Economy

50.0

Robustness

37.5

Productionizability

50.0

Modifiability

50.0

Comprehensibility

75.0

Responsibility

60.0

Action Summary

CRITICAL ACTIONS

5

IMPORTANT ACTIONS

3

NICE-TO-HAVE ACTIONS

6

Priority Actions

Critical

Availability (robustness)

Current gap: large | Blocks level: L1 | Required quality requirement: full

Scoring rationale: Full requirement is not yet demonstrated. Some answers were unknown and were scored conservatively.

Action: Define and meet availability SLA targets with monitored fallback behavior.

Fairness (responsibility)

Current gap: small | Blocks level: L1 | Required quality requirement: full

Scoring rationale: Minimal requirement is demonstrated, but full requirement is not yet demonstrated.

Action: Evaluate across relevant demographic groups and implement mitigation for harmful disparities.

Operability (modifiability)

Current gap: large | Blocks level: L1 | Required quality requirement: min

Scoring rationale: Neither minimal nor full requirement is currently demonstrated.

Action: Ensure versioned deployment and rollback.

Vulnerability (responsibility)

Current gap: large | Blocks level: L1 | Required quality requirement: full

Scoring rationale: Neither minimal nor full requirement is currently demonstrated. Some answers were unknown and were scored conservatively.

Action: Step 1: Implement baseline protections (input sanitization, secret handling, prompt injection checks). Step 2: Add multi-layered security controls including bot detection, red teaming exercises, PII redaction, jailbreak defenses, and security audits.

Testability (modifiability)

Current gap: large | Blocks level: L2 | Required quality requirement: min

Scoring rationale: Neither minimal nor full requirement is currently demonstrated.

Action: Reach at least 50% source code coverage.

Important

Accuracy (utility)

Current gap: small | Blocks level: L3 | Required quality requirement: full

Scoring rationale: Minimal requirement is demonstrated, but full requirement is not yet demonstrated.

Action: Expand to a 200+ example golden dataset with automated evals plus periodic human review.

Resilience (robustness)

Current gap: large | Blocks level: L3 | Required quality requirement: min

Scoring rationale: Neither minimal nor full requirement is currently demonstrated.

Action: Raise automated pipeline (e.g., prompt updates, RAG index refresh, continuous evaluation) success rate to at least 80% per quarter.

Traceability (comprehensibility)

Current gap: large | Blocks level: L3 | Required quality requirement: min

Scoring rationale: Neither minimal nor full requirement is currently demonstrated.

Action: Log prompt versions and core request metadata (e.g., timestamp, user ID, session ID, model version, model parameters).

Nice to Have

Adaptability (robustness)

Current gap: small | Blocks level: L4 | Required quality requirement: full

Scoring rationale: Minimal requirement is demonstrated, but full requirement is not yet demonstrated.

Action: Automate update loops using feedback and evaluation-gated releases.

Explainability (responsibility)

Current gap: small | Blocks level: L4 | Required quality requirement: full

Scoring rationale: Minimal requirement is demonstrated, but full requirement is not yet demonstrated.

Action: Provide user-facing transparency artifacts (citations, tool traces, confidence).

Repeatability (productionizability)

Current gap: small | Blocks level: L4 | Required quality requirement: full

Scoring rationale: Minimal requirement is demonstrated, but full requirement is not yet demonstrated.

Action: Fully automate update/evaluation flow with versioning.

Cost-Effectiveness (economy)

Current gap: large | Blocks level: L5 | Required quality requirement: full

Scoring rationale: Neither minimal nor full requirement is currently demonstrated.

Action: Step 1: Document and report all major cost components (API costs, storage, compute, human reviews, etc). Step 2: Demonstrate sustained positive ROI and establish budget guardrails.

Effectiveness (utility)

Current gap: small | Blocks level: L5 | Required quality requirement: full

Scoring rationale: Minimal requirement is demonstrated, but full requirement is not yet demonstrated.

Action: Conduct periodic A/B experiments (block-out) where you test against the previous control group or a baseline.

Monitoring (productionizability)

Current gap: small | Blocks level: L5 | Required quality requirement: full

Scoring rationale: Minimal requirement is demonstrated, but full requirement is not yet demonstrated.

Action: Add continuous automated output quality evaluation.

Gap Summary

SUB-CHARACTERISTIC	CATEGORY	GAP	BLOCKS LEVEL	PRIORITY
Availability	robustness	large	L1	critical
Fairness	responsibility	small	L1	critical
Operability	modifiability	large	L1	critical
Vulnerability	responsibility	large	L1	critical
Testability	modifiability	large	L2	critical
Accuracy	utility	small	L3	important
Resilience	robustness	large	L3	important
Traceability	comprehensibility	large	L3	important
Adaptability	robustness	small	L4	nice_to_have
Explainability	responsibility	small	L4	nice_to_have
Repeatability	productionizability	small	L4	nice_to_have
Cost-Effectiveness	economy	large	L5	nice_to_have
Effectiveness	utility	small	L5	nice_to_have
Monitoring	productionizability	small	L5	nice_to_have

Method: gap scores no=0, small=1, large=2; quality score uses denominator 2 x assessed sub-characteristics. Maturity level computed from Table 1 level gates encoded in configs.

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